



# TRULY INNOVATIVE ELECTRONICS

## INNOVATION UPDATES

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Electronics*

### Industry's first ultrasonic lens cleaning chipset

Texas Instruments (TI) has launched the industry's first semiconductors designed specifically with ultrasonic



lens cleaning (ULC) technology. The ULC chipset permits camera systems to autonomously detect and dispel dust, ice, and water through the use of minute vibrations. The chipset provides a streamlined and cost-effective solution for integrating ULC technology into various camera systems and sizes, offering broad application potential. This chipset facilitates the implementation of ULC technology in a minimal space, with a printed circuit board footprint under 25mm x 15mm. *Texas Instruments (TI)*  
<https://www.ti.com>

### World's smallest MEMS accelerometer

Bosch Sensortec has introduced the BMA580, an MEMS accelerometer. The accelerometer, with activity tracking and step counting, is suited for wearable technology. Compared to Bosch's



previous model, it occupies 76% less space and the profile has been reduced from 0.95mm to 0.55mm. The size of the accelerometer facilitates its placement on PCBs, and

its functionalities enable design engineers to incorporate them into portable devices. It uses bone conduction technology to sense the vibrations of the user's voice, activating the microphone from its sleep state when needed, thus saving power. The accelerometer is useful beyond wearables, for gesture recognition in toys, detecting falls in laptops and other electronics, and managing power in smartphones by detecting inactivity to initiate sleep mode.

*Bosch Sensortec*  
<https://www.bosch-sensortec.com>

### Nuclear battery to last 50 years without charging

Betavolt has developed a battery that can power smartphones for 50 years without recharging. This battery is con-



sidered the world's first miniaturised atomic energy system, with dimensions similar to a coin, measuring 15x15x5 cubic millimetres. The battery can deliver 100 microwatts of power at 3V, generating electricity from the decay of nickel-63. The battery is resistant to fire and impact, and can operate in temperatures from -60°C to 120°C. Its size allows for the connection of multiple units for more power. It can be used in aerospace, medical care, UAVs, micro-robots, microprocessors, advanced sensors, drones, and AI. It is designed to be safe, preventing fire or explosions, and emits no radiation, making it suitable for medical devices.

*Betavolt*  
<https://www.samsung.com>  
<https://imt-dubai.com/betavolt>

### Industry's first miniature SIP reed relay

Pickering Electronics has expanded its Series 104 reed relay line with a new model that features a 5kV stand-off and up to 1.5kV switching capability. The



new models can occupy the same printed circuit board (PCB) space

as one of the previously larger relays, with adequate separation between each component. The relays are suited for a variety of applications, including mixed signal semiconductor testing, cable tests, photovoltaic efficiency monitoring, electric vehicle and charging station testing, gas analysis in mining, medical electronics, in-circuit testing, high voltage instrumentation, and more.

*Pickering Electronics*  
<https://www.pickeringrelay.com>

### Low-power-consuming multi-core BLE chip

Renesas Electronics Corporation has launched the DA14592 Bluetooth Low Energy (LE) system-on-chip (SoC). The SoC is designed for low power con-



sumption and small size, featuring a multi-core architecture for Bluetooth LE applications. The SoC is aimed at uses such as connected medical devices, asset tracking, human interface devices, metering, point of sale (PoS) readers, and crowd-sourced location tracking. The DA14592 is notable for its cost-effectiveness, requiring few exter-